

Mathematics for Machine Learning: Complete Cheat Sheet

1. Linear Algebra

Vectors

Concepts:

- Vector: ordered list of numbers $[x_1, x_2, \dots, x_n]$
- Magnitude: $\|v\| = \sqrt{(x_1^2 + x_2^2 + \dots + x_n^2)}$
- Dot Product: $v \cdot w = \sum (v_i w_i)$
- Unit Vector: $v / \|v\|$

Real Applications:

- Word embeddings in NLP (word2vec, GloVe) represent words as vectors
- Image pixels as vectors for computer vision
- User preferences in recommendation systems
- Feature vectors in any ML model

Matrices

Concepts:

- Matrix: 2D array of numbers ($m \times n$)
- Transpose: A^T flips rows and columns
- Matrix Multiplication: $(AB)_{ij} = \sum_k A_{ik} B_{kj}$
- Identity Matrix: I (diagonal of 1s)
- Inverse: A^{-1} where $AA^{-1} = I$
- Determinant: $\det(A)$ - scalar value

Real Applications:

- Neural network weight matrices transform inputs
- Image transformations (rotation, scaling, shearing)
- Data transformation in preprocessing
- Covariance matrices for PCA

Eigenvalues & Eigenvectors

Concepts:

- $Av = \lambda v$ (v is eigenvector, λ is eigenvalue)
- Eigendecomposition: $A = Q\Lambda Q^{-1}$
- Captures directions of maximum variance

Real Applications:

- Principal Component Analysis (PCA) for dimensionality reduction
- Google PageRank algorithm
- Stability analysis in neural networks
- Face recognition (Eigenfaces)

Singular Value Decomposition (SVD)

Concepts:

- $A = U\Sigma V^T$
- U : left singular vectors, Σ : singular values, V : right singular vectors
- Works for any matrix (not just square)

Real Applications:

- Recommender systems (Netflix, Amazon)
- Latent Semantic Analysis in NLP
- Image compression
- Noise reduction in data

Norms

Concepts:

- L1 norm: $\|x\|_1 = \sum |x_i|$
- L2 norm: $\|x\|_2 = \sqrt{\sum x_i^2}$
- L ∞ norm: $\|x\|_\infty = \max(|x_i|)$

Real Applications:

- L1 regularization (Lasso) for feature selection
 - L2 regularization (Ridge) to prevent overfitting
 - Distance metrics in clustering (K-means)
 - Loss function components
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2. Calculus

Derivatives

Concepts:

- $f'(x) = \lim_{h \rightarrow 0} (f(x+h) - f(x))/h$
- Chain Rule: $(f \circ g)'(x) = f'(g(x)) \cdot g'(x)$
- Partial Derivatives: $\partial f / \partial x$ (derivative with respect to one variable)

Real Applications:

- Gradient descent optimization
- Backpropagation in neural networks
- Finding optimal parameters
- Sensitivity analysis

Gradients

Concepts:

- $\nabla f = [\partial f / \partial x_1, \partial f / \partial x_2, \dots, \partial f / \partial x_n]$
- Direction of steepest ascent
- Points perpendicular to level curves

Real Applications:

- Training all ML models (gradient descent)
- Computer vision edge detection
- Optimization algorithms (Adam, RMSprop)
- Neural network weight updates

Jacobian Matrix

Concepts:

- Matrix of all first-order partial derivatives
- $J = [\partial f_i / \partial x_j]$
- Maps how vector-valued function changes

Real Applications:

- Neural network layer transformations

- Robotics (velocity transformations)
- Automatic differentiation frameworks
- Sensitivity of model outputs

Hessian Matrix

Concepts:

- Matrix of second-order partial derivatives
- $H = [\partial^2 f / \partial x_i \partial x_j]$
- Indicates curvature

Real Applications:

- Second-order optimization methods
- Identifying saddle points in loss landscapes
- Newton's method for optimization
- Understanding model convergence

Integration

Concepts:

- $\int f(x) dx$ (area under curve)
- Definite integral: $\int_a^b f(x) dx$
- Fundamental Theorem: $\int f'(x) dx = f(x) + C$

Real Applications:

- Probability density functions (computing probabilities)
 - Expected values in reinforcement learning
 - Bayesian inference
 - Monte Carlo methods
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3. Probability & Statistics

Probability Basics

Concepts:

- $P(A)$: probability of event A, range $[0,1]$

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A|B) = P(A \cap B)/P(B)$ (conditional probability)
- Bayes' Theorem: $P(A|B) = P(B|A)P(A)/P(B)$

Real Applications:

- Spam filtering (Naive Bayes)
- Medical diagnosis systems
- Weather prediction models
- Document classification

Random Variables

Concepts:

- Discrete: finite/countable outcomes
- Continuous: infinite outcomes in range
- Expected Value: $E[X] = \sum x \cdot P(X=x)$
- Variance: $\text{Var}(X) = E[(X-\mu)^2]$

Real Applications:

- Model uncertainty quantification
- A/B testing analysis
- Financial risk modeling
- Sensor noise modeling

Probability Distributions

Discrete:

- Bernoulli: single yes/no trial
- Binomial: n independent trials
- Poisson: count of events in interval

Continuous:

- Uniform: all values equally likely
- Normal/Gaussian: $N(\mu, \sigma^2)$, bell curve
- Exponential: time between events

Real Applications:

- Logistic regression (Bernoulli for binary classification)
- Image pixel intensities (Gaussian noise)
- Natural language word frequencies (Poisson)
- Generative Adversarial Networks (various distributions)

Central Limit Theorem

Concepts:

- Sum of independent random variables $\rightarrow$ Normal distribution
- Holds regardless of original distribution
- $n \geq 30$ typically sufficient

Real Applications:

- Confidence intervals for model predictions
- Bootstrap sampling methods
- A/B test validity
- Error analysis

Maximum Likelihood Estimation (MLE)

Concepts:

- Find parameters that maximize $P(\text{data}|\text{parameters})$
- $L(\theta) = \prod P(x_i|\theta)$
- Often use log-likelihood: $\log L(\theta) = \sum \log P(x_i|\theta)$

Real Applications:

- Training logistic regression
- Fitting Gaussian Mixture Models
- Parameter estimation in any probabilistic model
- Neural network optimization (cross-entropy loss)

Hypothesis Testing

Concepts:

- Null hypothesis H_0 vs alternative H_1
- p-value: probability of observing data if H_0 true
- Significance level α (typically 0.05)

- Type I error (false positive), Type II error (false negative)

Real Applications:

- A/B testing for product features
- Model comparison (is new model better?)
- Feature selection significance
- Clinical trial analysis

Covariance & Correlation

Concepts:

- $\text{Cov}(X, Y) = E[(X - \mu_x)(Y - \mu_y)]$
- Correlation: $\rho = \text{Cov}(X, Y) / (\sigma_x \sigma_y)$, range $[-1, 1]$
- Covariance Matrix: $\Sigma_{ij} = \text{Cov}(X_i, X_j)$

Real Applications:

- Feature correlation analysis
 - Portfolio optimization
 - Multivariate Gaussian distributions
 - Dimensionality reduction (PCA)
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4. Optimization

Gradient Descent

Concepts:

- $\theta_{new} = \theta_{old} - \alpha \nabla f(\theta)$
- α : learning rate
- Iteratively move toward minimum
- Batch, Mini-batch, Stochastic variants

Real Applications:

- Training all neural networks
- Linear regression optimization
- Logistic regression training
- Deep learning model optimization

Convex Optimization

Concepts:

- Convex function: $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$
- Single global minimum
- Gradient descent guaranteed to converge

Real Applications:

- Support Vector Machines (SVM)
- Linear regression (closed-form solution)
- Lasso and Ridge regression
- Many traditional ML algorithms

Lagrange Multipliers

Concepts:

- Optimize $f(x)$ subject to $g(x) = 0$
- $\nabla f = \lambda \nabla g$ at optimum
- Handles constrained optimization

Real Applications:

- SVM margin optimization
- Constrained neural network training
- Resource allocation problems
- Portfolio optimization with constraints

Adam, RMSprop, Momentum

Concepts:

- Momentum: accumulates velocity
- RMSprop: adaptive learning rate
- Adam: combines momentum + RMSprop
- Faster convergence than vanilla gradient descent

Real Applications:

- Default optimizer for deep learning

- Training large language models
 - Computer vision models (ResNet, YOLO)
 - Faster training of complex models
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5. Information Theory

Entropy

Concepts:

- $H(X) = -\sum P(x)\log P(x)$
- Measures uncertainty/information content
- Higher entropy = more uncertain

Real Applications:

- Decision tree splitting criteria
- Feature selection
- Compression algorithms
- Measuring model uncertainty

Cross-Entropy

Concepts:

- $H(P,Q) = -\sum P(x)\log Q(x)$
- Measures difference between distributions
- Used as loss function

Real Applications:

- Classification loss function
- Training neural networks
- Natural language processing models
- Image classification (softmax + cross-entropy)

KL Divergence

Concepts:

- $D_{KL}(P\|Q) = \sum P(x)\log(P(x)/Q(x))$

- Measures how Q differs from P
- Not symmetric: $D_{KL}(P||Q) \neq D_{KL}(Q||P)$

Real Applications:

- Variational Autoencoders (VAE)
- Generative models evaluation
- Bayesian inference
- Model comparison

Mutual Information

Concepts:

- $I(X;Y) = H(X) + H(Y) - H(X,Y)$
- Measures dependence between variables
- $I(X;Y) = 0$ if X,Y independent

Real Applications:

- Feature selection algorithms
 - Image registration
 - Clustering evaluation
 - Causal inference
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6. Advanced Topics

Taylor Series

Concepts:

- $f(x) \approx f(a) + f'(a)(x-a) + \frac{f''(a)(x-a)^2}{2!} + \dots$
- Approximates functions as polynomials
- First-order: linear approximation

Real Applications:

- Activation function approximations
- Newton's method optimization
- Numerical gradient computation
- Understanding neural network training dynamics

Fourier Transform

Concepts:

- Decomposes signal into frequency components
- $F(\omega) = \int f(t)e^{-i\omega t} dt$
- Converts time domain $\leftrightarrow$ frequency domain

Real Applications:

- Audio processing (speech recognition)
- Image filtering and compression
- Signal denoising
- Convolutional Neural Networks (frequency domain convolutions)

Graph Theory Basics

Concepts:

- Graph: $G = (V, E)$ vertices and edges
- Adjacency matrix A : $A_{ij} = 1$ if edge exists
- Degree, paths, connectivity
- Laplacian matrix

Real Applications:

- Graph Neural Networks (GNN)
- Social network analysis
- Knowledge graphs
- Molecular structure prediction

Distance Metrics

Concepts:

- Euclidean: $d(x, y) = \sqrt{\sum (x_i - y_i)^2}$
- Manhattan: $d(x, y) = \sum |x_i - y_i|$
- Cosine Similarity: $\cos(\theta) = (x \cdot y) / (\|x\| \|y\|)$
- Hamming: count of differing positions

Real Applications:

- K-Nearest Neighbors (KNN)
- Clustering algorithms
- Similarity search
- Recommendation systems

Markov Chains

Concepts:

- State transitions with probabilities
- $P(X_{n+1}|X_n, X_{n-1}, \dots) = P(X_{n+1}|X_n)$
- Transition matrix, stationary distribution

Real Applications:

- Hidden Markov Models (speech recognition)
 - PageRank algorithm
 - Text generation models
 - Reinforcement learning (MDPs)
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7. Practical ML Math Applications

Neural Network Forward Pass

$z = Wx + b$ (linear transformation)

$a = \sigma(z)$ (activation function)

Repeat for each layer

Math Used: Matrix multiplication, vector addition, function composition

Backpropagation

$\partial L / \partial W = \partial L / \partial a \cdot \partial a / \partial z \cdot \partial z / \partial W$ (chain rule)

$W := W - \alpha (\partial L / \partial W)$ (gradient descent)

Math Used: Chain rule, partial derivatives, gradient descent

Principal Component Analysis (PCA)

1. Center data: $\tilde{X} = X - \text{mean}(X)$

2. Compute covariance: $C = (\tilde{X}^T \tilde{X}) / n$

3. Find eigenvectors of C
4. Project: $X_{\text{new}} = \tilde{X} \cdot V_k$

Math Used: Eigenvalues, eigenvectors, matrix multiplication

Linear Regression

$$\hat{y} = X\beta$$
$$\beta = (X^T X)^{-1} X^T y \text{ (closed form)}$$
$$\text{Loss: } L = \|y - X\beta\|^2$$

Math Used: Matrix inverse, linear algebra, L2 norm

Logistic Regression

$$\sigma(z) = 1/(1+e^{(-z)})$$
$$L = -[y \cdot \log(\hat{y}) + (1-y) \cdot \log(1-\hat{y})]$$

Math Used: Sigmoid function, cross-entropy, gradient descent

Support Vector Machines

$$\text{Maximize margin: } 2/\|w\|$$
$$\text{Subject to: } y_i(w \cdot x_i + b) \geq 1$$

Math Used: Lagrange multipliers, convex optimization, dot products

K-Means Clustering

1. Assign: $c_i = \text{argmin}_k \|x_i - \mu_k\|^2$
2. Update: $\mu_k = \text{mean}(\text{points in cluster } k)$

Math Used: Euclidean distance, mean calculation, iterative optimization

Gaussian Mixture Models

$$P(x) = \sum_k \pi_k N(x|\mu_k, \Sigma_k)$$

EM algorithm for parameter estimation

Math Used: Multivariate Gaussian, probability, expectation-maximization

8. Key Formulas Quick Reference

Activation Functions:

- Sigmoid: $\sigma(x) = 1/(1+e^{(-x)})$
- Tanh: $\tanh(x) = (e^x - e^{(-x)})/(e^x + e^{(-x)})$
- ReLU: $\max(0, x)$
- Softmax: $\sigma(x)_i = e^{(x_i)} / \sum_j e^{(x_j)}$

Loss Functions:

- MSE: $(1/n) \sum (y - \hat{y})^2$
- Cross-Entropy: $-\sum y \cdot \log(\hat{y})$
- Hinge Loss: $\max(0, 1 - y \cdot \hat{y})$

Regularization:

- L1: $\lambda \sum |w_i|$
- L2: $\lambda \sum w_i^2$
- Elastic Net: $\lambda_1 \sum |w_i| + \lambda_2 \sum w_i^2$

Metrics:

- Accuracy: $(TP+TN)/(TP+TN+FP+FN)$
 - Precision: $TP/(TP+FP)$
 - Recall: $TP/(TP+FN)$
 - F1: $2 \cdot (\text{Precision} \cdot \text{Recall}) / (\text{Precision} + \text{Recall})$
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9. Study Path Recommendation

Foundation (2-3 weeks):

1. Linear algebra: vectors, matrices, operations
2. Basic calculus: derivatives, chain rule
3. Probability basics

Intermediate (3-4 weeks):

1. Matrix decompositions (SVD, eigenvalues)
2. Gradients, Jacobians

3. Probability distributions, MLE

Advanced (4-6 weeks):

1. Optimization algorithms
2. Information theory
3. Apply to real ML problems

Practice:

- Implement algorithms from scratch (numpy)
 - Derive backpropagation manually
 - Work through ML papers with focus on math
 - Use libraries (scikit-learn, PyTorch) to see math in action
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10. Resources for Deep Dive

Books:

- "Mathematics for Machine Learning" by Deisenroth, Faisal, Ong
- "Pattern Recognition and Machine Learning" by Bishop
- "Deep Learning" by Goodfellow, Bengio, Courville

Online:

- 3Blue1Brown (YouTube) - visual explanations
- Khan Academy - fundamentals
- FastAI - practical applications
- MIT OpenCourseWare - rigorous foundations